MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
V SEMESTER ELECTRONICS ELECTIVE PRACTICAL EXAMINATION
(2024 ADMISION ONWARDS)

MG5DSEECT310- Robotics and Industrial Automation

PRACTICAL EXAMINATION

Duration: 2 Hours **Maximum Marks: 35**

Instructions:
- Demonstrate the given task. (Any one of the tasks listed below)
- Viva session (Minimum 5 questions from the Practical Module)
- Lab Report (certified)

Evaluation Criteria:
- Viva: 10 marks
- Lab Report: 5 marks
- Demonstration: 20 marks

For Practical / Simulation Use (OpenPLC Editor, i- TRiLOGI, WPLSoft, Do-more Designer, plcsimulator.online or any other).

1. Basic ON/OFF Control: Use a switch to control an output (e.g., a lamp) using PLC.
2. Toggle Operation: Implement a toggle switch to alternate between two outputs.
3. Timer Functionality: Use timers to control the ON/OFF duration of an output.
4. Latching Circuit: Create a latch/unlatch mechanism to maintain output state.
5. Logic Gates Implementation: Use PLC programming to simulate AND, OR, NOT logic functions.
6. Motor Control: Control the direction and speed of a motor using PLC.
7. Traffic Light Simulation: Simulate a traffic light system with different timing sequences.
8. Temperature Control: Control a heating or cooling system based on temperature sensor inputs.
9. Water Level Monitoring: Use sensors to monitor and control water levels in a tank.
10. Conveyor Belt Control: Control the operation and speed of a conveyor belt using PLC.
11. Alarm System: Create an alarm system based on sensor inputs or specific conditions.
12. Robotic Arm Control: Basic control of a robotic arm using PLC
13. Robotic Application: Robotic arm pick-and-place tasks using PLC